

Probability and Curve Fitting

Let's face it: There are too many variables here, all of which have similar names and definitions. There are 3 different standard deviations in play! You'll get very familiar with all of this as you work through Homework 1. Hopefully this helps:

For a single measurement...

The “Gaussian” or “Normal” Probability Density Function

$$p(y) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(y-\mu)^2}{2\sigma^2}}$$

For any probability density function $p(y)$, the quantity $p(y)dy$ is the probability that a number drawn at random from the given distribution will be between y and $y + dy$. If you draw a million numbers from a distribution defined by $p(y)$, a histogram of those numbers will have the same shape as $p(y)$.

The Gaussian or normal distribution is special: the **central limit theorem** says that when many different random variables contribute additively to an outcome, the outcome has a Gaussian probability distribution. *This is true even when the individual contributing random variables do not have Gaussian distributions!*

μ : True Mean

The peak and center of a Gaussian distribution that you are drawing from. You often can't know the true mean. If you had a perfect model for your data, the model would predict the true mean. The definition of expectation value says that for the distribution above, $\langle y \rangle = \mu$.

σ : True Standard Deviation

The *one-sided width* of a Gaussian distribution that you are drawing from. 68% of the time, you will draw a number within $\pm 1\sigma$ of the true mean μ .

$$\int_{-\sigma}^{+\sigma} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(y-\mu)^2}{2\sigma^2}} dy \approx 0.68$$

You often can't know the true standard deviation. Even a model of your system can't predict this because it has to do with the net effect of all measurement errors.

σ^2 : True Variance

$$\langle (y - \mu)^2 \rangle = \sigma^2$$

Variances are just squares of standard deviations. These often have nicer mathematical properties than the standard deviation. For example, if you draw several random numbers and sum them together, the variance of the sum is the sum of the variances of each one. This is not true for the standard deviations. Physicists report σ rather than σ^2 because values of σ are more directly comparable to values of y . After all, σ of a thing has the same units as the thing itself.

For a sample of $N \approx 5$ measurements with the same settings...

We'll make each measurement $N \approx 5$ times. For example, at *each* knob position, we'll measure the intensity of light $N \approx 5$ times to get $N \approx 5$ measurements: $\{y_1, y_2, \dots, y_N\}$.

$\bar{y}$: Sample Mean

The average of the $N \approx 5$ samples:

$$\bar{y} = (y_1 + y_2 + \dots + y_N) / N = \frac{1}{N} \sum_{n=1}^N y_n.$$

This is an *unbiased estimate of the true mean* μ : given $N \approx 5$ measurements, forming their mean is the best you can do—it won't overestimate or underestimate μ .

This kind of sample mean typically becomes the y value of a *single data point* in a plot. (The point's error bars are described below.)

Math note: If you draw N samples and calculate their mean, and you do that procedure a bazillion times, the mean of these *sample means* will converge on the true mean: $\langle \bar{y} \rangle = \mu$.

s^2 : Sample Variance

$$s^2 = \frac{1}{N-1} \sum_{n=1}^N (y_n - \bar{y})^2$$

This is an *unbiased estimate of the true variance* σ^2 in the same sense as above: given $N \approx 5$ measurements, computing s^2 is the best you can do—on average, it won't overestimate or underestimate σ^2 .

Math note: If you draw $N \approx 5$ samples and calculate their s^2 a bazillion times, the mean of these bazillion sample variances will converge on the true variance: $\langle s^2 \rangle = \sigma^2$.

s : Sample Standard Deviation

Square root of s^2 from above. Unfortunately, this is *not* an unbiased estimator of σ . If you do this procure for a bazillion groups of $N \approx 5$ samples, the average of these bazillion s calculations won't converge to σ . It'll be a little too small on average, even if you correctly use $N-1$ in the formula above: $\langle s \rangle < \sigma$. There's no simple unbiased estimator of σ itself. That's mostly OK; we talk about s a lot because we like its units, but all the important statistical analysis is actually based on s^2 anyway.

$\sigma_{\bar{y}}$ or $s_{\bar{y}}$: Sample Standard Deviation of the Mean; (or “Standard Error of the Mean (SEM)”)

$$\sigma_{\bar{y}} = \frac{\sigma}{\sqrt{N}} \quad \text{or estimated from the sample standard deviation as} \quad s_{\bar{y}} = \frac{s}{\sqrt{N}}$$

A mean of N samples (a sample mean $\bar{y}$) gets closer to the true mean μ as you increase your sample size N (say from 5, to 10, to 100). Increasing N gives you get a better estimate of the true mean μ . How much better? Better (narrower) by $\sqrt{N}$.

If you draw N samples and form their sample mean $\bar{y}$, and you do that procedure a bazillion times to make a histogram, the one-sided width of that histogram would be $\sigma_{\bar{y}}$. Around 68% of sample means $\bar{y}$ will be within $\pm 1\sigma_{\bar{y}}$ of the true mean μ .

It's as if $\bar{y}$ were drawn from a normal distribution of mean μ and standard deviation $\sigma_{\bar{y}}$.

This $\sigma_{\bar{y}}$, or really our best estimate $s_{\bar{y}}$ of it, becomes the *one-sided size of the error bar* of a data point in a plot.

Curve Fitting

You have a model where you can turn a knob to some value x and predict what you'd expect for a measurement of, say, the intensity y of light there. The model has some functional form $f(x)$. For a linear model, $f(x) = ax + b$. The goal is to decide how well the functional form matches the data, and to find the “best-fit” values of the parameters like a and b .

You'll sweep through, say, $M \approx 9$ knob positions $N \approx 5$ times. You'll end up with $N \approx 5$ independent measurements for each of $M \approx 9$ knob positions. This is $NM \approx 45$ numbers. The discussion here assumes the same number of trials N at each knob position, but if this is not true the essentials of the analysis still hold, only with even more subscripts (N becomes N_m).

For each of the $M \approx 9$ independent-variable positions $\{x_1, x_2, \dots, x_M\}$, you will compute:

- The sample mean $\bar{y}_m$ (each of the $M \approx 9$ sample means is the average of $N \approx 5$ independent measurements with the knob at that particular x_m). These $M \approx 9$ means are your $M \approx 9$ data points on the plot.
- The standard error of the mean $\sigma_{\bar{y}_m}$ estimated from data to be $s_{\bar{y}_m}$. These are the error bars attached to each point. They go $+s_{\bar{y}_m}$ above the point to $-s_{\bar{y}_m}$ below.

At each knob position x_m , we assume individual measurements (single trials) are drawn at random from a normal distribution centered on $f(x_m) = ax_m + b$:

$$p(y_m) = \frac{1}{\sqrt{2\pi\sigma_m^2}} e^{-\frac{(y_m - f(x_m))^2}{2\sigma_m^2}}.$$

The variance σ_m^2 of each measurement (each trial) is determined by a combination of independent errors that lead to a spread in measurements at each particular knob position x_m . We don't know σ_m^2 , but we can estimate it as the sample variance s_m^2 .

Each sample mean $\bar{y}_m$ (each data point on the plot) is drawn from a normal distribution centered on the same mean value, but with a narrower width $\sigma_{\bar{y}_m} = \frac{\sigma_m}{\sqrt{N}}$. Again, our best estimate of this is $s_{\bar{y}_m} = \frac{s_m}{\sqrt{N}}$, so each data point has probability distribution

$$p(\bar{y}_m) = \frac{1}{\sqrt{2\pi s_{\bar{y}_m}^2}} e^{-\frac{(\bar{y}_m - f(x_m))^2}{2s_{\bar{y}_m}^2}}.$$

If we plot the data points with error bars on top of the model line or curve, around 68% of the points should be within the two-sided error bar of the curve if the model is a good description of the measured phenomenon and our error bars were estimated well.

We assume that each data point is statistically independent, meaning the joint probability distribution for all M data points is:

$$p(\bar{y}_1, \bar{y}_2, \dots, \bar{y}_M) = p(\bar{y}_1)p(\bar{y}_2)\dots p(\bar{y}_M) = \left[\prod_{m=1}^M \left(\frac{1}{\sqrt{2\pi s_{\bar{y}_m}^2}} \right) \right] e^{-\chi^2/2},$$

where

$$\chi^2 \equiv \sum_{m=1}^M \left(\frac{\bar{y}_m - f(x_m)}{s_{\bar{y}_m}} \right)^2$$

is a measure of how far away the points are from the curve, compared to their own standard deviations.

The best fit is where the adjustable parameters (a and b in the case of a straight-line function) maximize the joint probability of our data points. (This is called the *principle of maximum likelihood* for defining the meaning of a “best” estimator or parameter.) Since the giant product in square brackets does not depend on the fit parameters, the maximum joint probability happens when the parameters minimize χ^2 . For a general function with lots of adjustable parameters, a computer can search for the parameter values that minimize χ^2 and explore the width of the minimum as a function of the parameters to estimate the uncertainties (such as σ_a and σ_b) of the parameter values. For the special case of a linear fit where $f(x) = ax + b$, you can actually take the derivatives of χ^2 with respect to a and b , set them each to 0, and solve exactly for the best a and b using linear algebra. The 1-sigma width σ_a and σ_b can then be computed by propagation of error.

χ^2 of a Fit, the χ^2 Distribution, and the Probability to Exceed

Once you have the best-fit parameters, found by minimizing χ^2 (shifting the fit line as close to the data points as possible), you can look at the value of χ^2 itself for those parameters:

$$\chi_{\text{fit}}^2 \equiv \sum_{m=1}^M \left(\frac{\bar{y}_m - y_m^{\text{fit}}}{s_{\bar{y}_m}} \right)^2$$

Each term in the sum is the squared number of standard deviations between the data point and the fit line. In other words, it's the vertical distance squared from the point to the line, in units of standard deviations (or equivalently in numbers of error bars).

One could also consider the χ^2 for the true model, assuming you know the true parameters

$$\chi_{\text{true}}^2 \equiv \sum_{m=1}^M \left(\frac{\bar{y}_m - y_m^{\text{true}}}{\sigma_{\bar{y}_m}} \right)^2.$$

For the true model, each point should be, on average, one standard deviation away. So χ_{true}^2 should be approximately M , the number of data points; if you look at a distribution of χ_{true}^2 , it should have a mean of M . This makes $\chi_{\text{true}}^2/M \approx 1$.

The distribution of χ_{true}^2 itself is called “the χ^2 distribution with M degrees of freedom.”¹ Because you’ve divided by the standard error in each term, this is the same distribution as the sum of the squares of M standard normal random variables (each with $\mu = 0$ and $\sigma = 1$).

However, since you don’t know the true model, and since you used the data itself to find the best-fit parameters that minimized χ^2 , your fit is typically going to be a little closer to your particular set of points than the true function would be. (Imagine fitting M points with a polynomial of degree M : It would go exactly through all of the points!)

We’ll call the number of fit parameters p . For a line with a slope and intercept that can both be adjusted to match the line to the data, $p = 2$. It can be shown that *each fit parameter effectively removes one degree of freedom*: The distribution of χ_{fit}^2 that you’d get by doing a bazillion fits to a bazillion data sets is the “ χ^2 distribution with $\nu = M - p$ degrees of freedom.” It’s the distribution that you’d get if you summed only ν samples drawn from the standard normal distribution.

You can look up how likely your particular value of χ_{fit}^2 is, given ν . In particular we ask, “What’s the probability that I’d find a χ_{fit}^2 value *greater* than the one I actually found if I did the experiment again using the same number of data points with the same uncertainties?” This is called the **Probability to Exceed (PTE)**, sometimes written $P_{>}$. If you repeat the whole experiment and fit many times, the distribution of $P_{>}$ values should be flat from 0 to 1. A good fit has a PTE between 5% and 95%, which should happen 90% of the time. An OK fit has a PTE between 1% and 99%, which should happen 98% of the time.

Just like the denominator of the sample variance needed $N - 1$ instead of N , χ_{fit}^2 has a mean of $\nu = M - p$ instead of M . This means that

$$\frac{\chi_{\text{fit}}^2}{\nu} = \frac{\chi_{\text{fit}}^2}{M - p} \approx 1 \quad \text{just like} \quad \frac{\chi_{\text{true}}^2}{M} \approx 1$$

¹https://en.wikipedia.org/wiki/Chi-square_distribution

We look to χ_{fit}^2/ν as a partial measure of “goodness of fit.” For a good fit, this is around 1.

If $\chi_{\text{fit}}^2/\nu \gg 1$, it means that the points were too many standard errors away from the fit. Either the error bars were too small or the function isn’t a good model for the data.

If $\chi_{\text{fit}}^2/\nu \ll 1$, it means that the points are too close to the fit. Either the error bars were too big or you used too many (often non-physical) parameters in the fit (called overfitting).

Rather than trying to be quantitative about the meaning of $\gg$ or $\ll$ in these guidelines, you should rely more on the value of $P_{>}$ when a quantitative judgement is required.